

LegalAid Call Abandonment Final Presentation

Simple EDA Team 1

A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

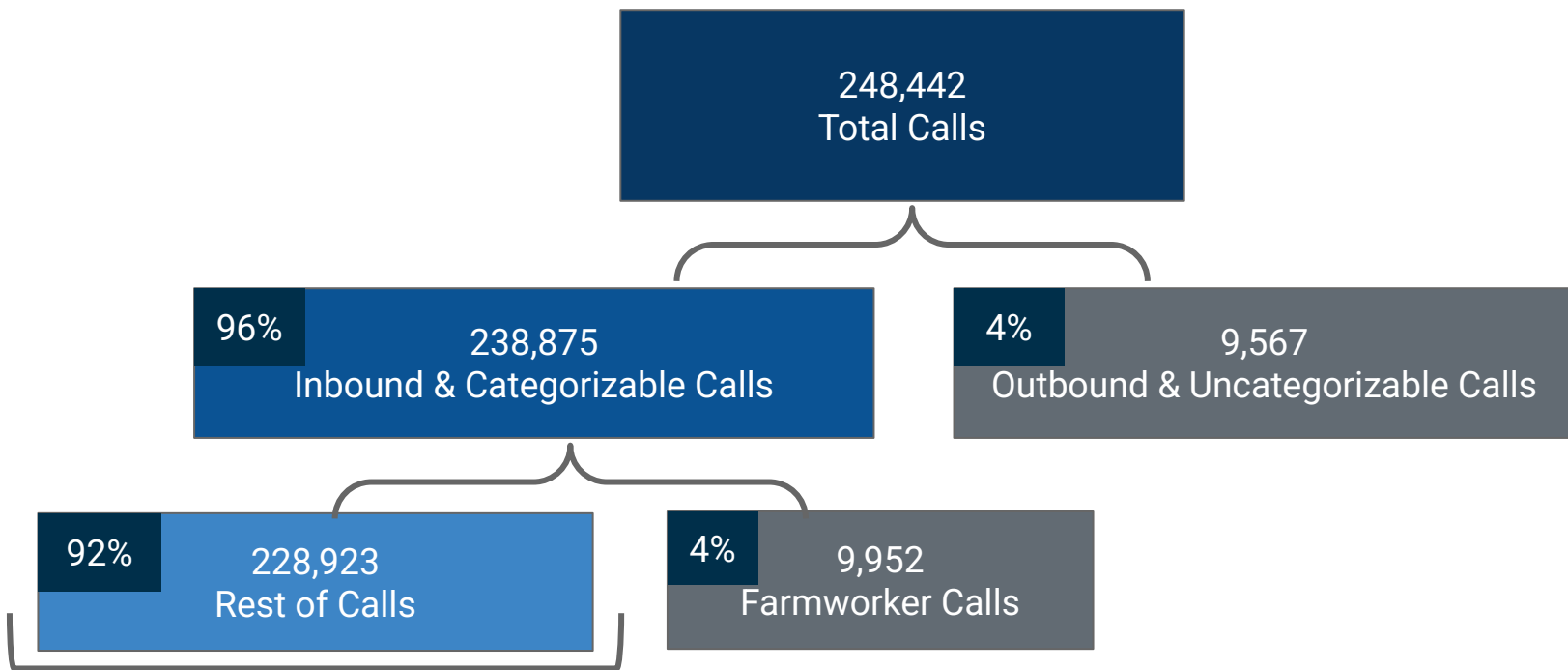
Introduction

Objective: Find overall proportion of abandoned calls, as well as proportion of abandonment by menu

Process:

- 1) Tag every call to a specific *menu* and *last activity* based on where they ended
- 2) Tag every subgroup of menu and last activity as either *abandoned* or *not abandoned*
- 3) Use these results to calculate the % of abandoned calls, both in total as well as per menu

Total Analysis Figures



Abandonment Calculations

Abandonment Calculations

Menu Name	Total Calls	% of All Calls	Abandoned Calls	Abandonment Rate
Benefits	7,210	3.15%	429	5.95%
Employment	4,564	1.99%	289	6.33%
Family	22,753	9.94%	2,242	9.85%
HIV	1,603	0.70%	767	47.85%
Housing	16,492	7.20%	3,837	23.27%
Immigration	3,516	1.54%	696	19.80%
Legal	27,715	12.11%	13,793	49.77%
Main	119,927	52.39%	40,657	33.90%
Other Legal	12,204	5.33%	6,830	55.97%
Pre-Legal Seniors	12,939	5.65%	4,425	34.20%
Totals	228,923		73,965	32.31%

Preprocessing Method – Python

- 1) New dataframe - one row per call
- 2) Per call - I found both the last menu each call passed through (of the 11 relevant menus) and I found the last 'Activity Name'
- 3) Aggregate, grouping by last menu and last activity name, finding counts for each of these subgroups
- 4) I copied my results to Google Sheets to compare with Meera's independent numbers using her method in R

Next: Each unique combination of last menu & last activity can be used to tag a subgroup of calls as abandoned or not in a MECE (mutually exclusive & collectively exhaustive) manner, powering our analysis

Menu	Last Activity Name	Count	Abandoned?
Farmworker Main Number Telephony EP	FarmworkerMainMenu	9640	
Farmworker Main Number Telephony EP	LanguageSelectionMenu	156	
Farmworker Main Number Telephony EP	MigrantVoicemailTransfer	67	
Farmworker Main Number Telephony EP	DisconnectContact1	44	
Farmworker Main Number Telephony EP	ClosedMenu	29	
Farmworker Main Number Telephony EP	ClinicVoicemailTransfer	7	
Farmworker Main Number Telephony EP	StaffDirectorySpanishTransfer	5	
Farmworker Main Number Telephony EP	StaffDirectoryEnglishTransfer	3	
Farmworker Main Number Telephony EP	AddressFaxHoursMenu	1	
Legal Benefits Menu Telephony EP	ClosedQueueMenu	4674	
Legal Benefits Menu Telephony EP	ClinicVoicemailTransfer	818	

```
df_final['Menu'].value_counts()
```

```
Main Number Telephony EP      120093
Legal Menu Telephony EP        27715
Legal Family Menu Telephony EP  22753
Legal Housing Menu Telephony EP 16492
Pre-Legal Menu Seniors Menu Telephony EP 12944
Other Legal Menu Telephony EP  12204
Farmworker Main Number Telephony EP 9953
Legal Benefits Menu Telephony EP  7210
Legal Employment Menu Telephony EP 4564
Legal Immigration Menu Telephony EP 3516
Legal HIV Menu Telephony EP     1603
```

Preprocessing Method – R (Meera)

- Forward-filled *activity name* and *ep name* to impute NA values
- Calculated the *ep name* journey of each call
 - Created variables representing **each preceding distinct ep name** per call
 - Last row of every call contains every ep it went through
- Sliced out the last row of every call
- Counted by *activity name*, *ep name*, and *all preceding ep names* to get the **count of each specific call journey**
- Created an *analysis* variable that extracts the most recent ep name within the specific menu options:
 - Farmworker, Benefits, Employment, Family, HIV, Housing, Immigration, Legal, Main, Other Legal, Pre-Legal Seniors, Intake Outdial*
- Counted by *analysis* and *activity name* → same results as Adam

contact_session_id	ep_name
4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Pre-Legal Menu Seniors Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Pre-Legal Menu Seniors Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Legal Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Legal Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP

last_distinct_before_last_ep	last_distinct_before_last_ep_prev	last_distinct_before_last_ep_prev2	last_distinct_before_last_ep_prev3
Closed Hours-Holidays Menu Telephony EP	Legal Menu Telephony EP	Pre-Legal Menu Seniors Menu Telephony EP	Main Number Telephony EP

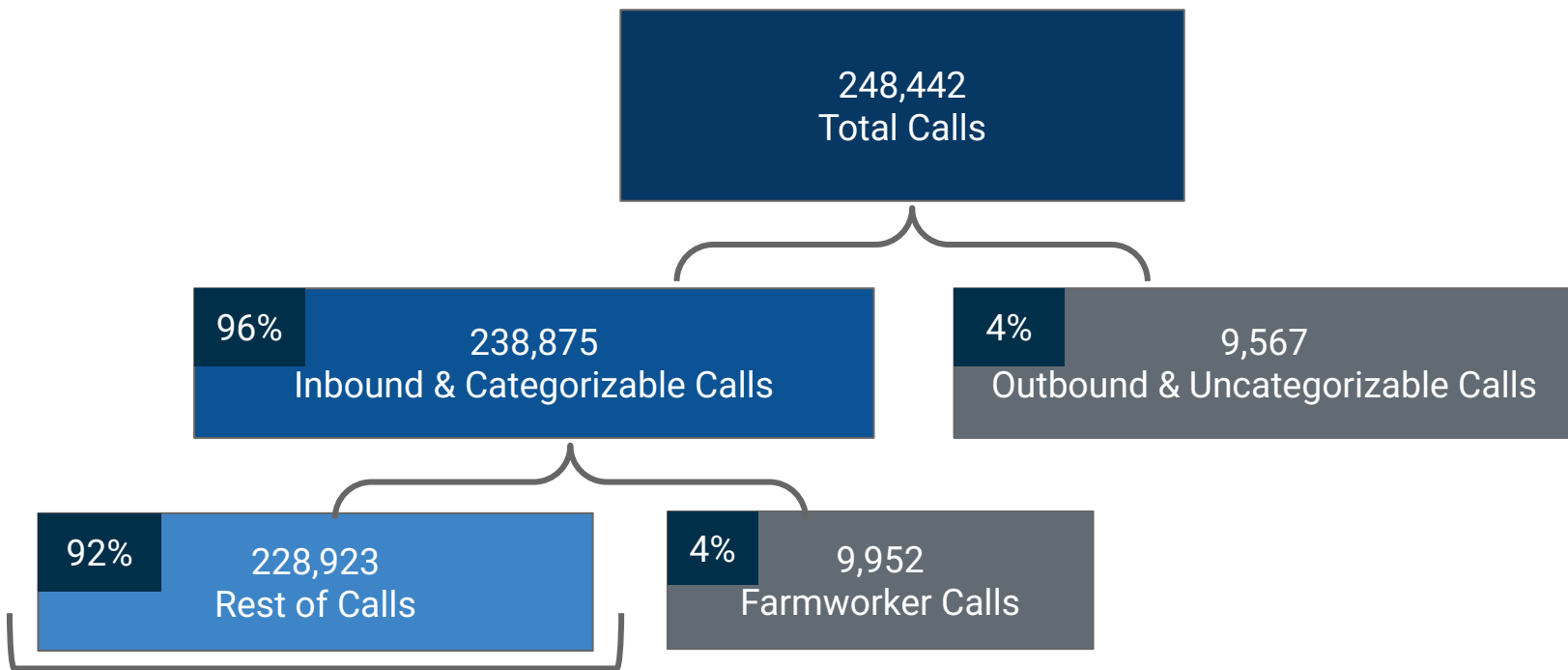
Outbound & Low Information Calls – Meera

1. Created a separate dataset for all calls with NA values for *activity name* throughout the whole call
2. Used the *termination reason* or *queue name* to categorize them through a “case when” statement
3. Counted by *ep name* and *activity name*
4. Added to spreadsheet to complete call categorization but excluded from abandonment calculations

ep_name	activity_name	n
Farmworker Main Number Telephony EP	No Activity	1
Intake Outdial EP	Intake Outdial Queue	2
Intake Outdial EP	System Error	3
Main Number Telephony EP	Customer Left	9
Main Number Telephony EP	NO_ANSWER_FROM_CUSTOMER	108
Main Number Telephony EP	No Activity	48
Main Number Telephony EP	System Error	1
Pre-Legal Menu Seniors Menu Telephony EP	NO_ANSWER_FROM_CUSTOMER	3
Pre-Legal Menu Seniors Menu Telephony EP	No Activity	2

No Activity Example	ep_name	flow_name	activity_name	timestamp	queue_name	agent_name	termination_reason
09be3dab-c9ed-4f0d-a123-b21942cd430e	Main Number Telephony EP	NA	NA	2024-12-16 13:18:55	NA	NA	NA
09be3dab-c9ed-4f0d-a123-b21942cd430e	Main Number Telephony EP	NA	NA	2024-12-16 13:18:55	NA	NA	NA

Total Analysis Figures



Abandonment Calculations

Spreadsheet Fill Process

From Code

Manual Fill

MENU	LAST ACTIVITY NAME	COUNT	Abandoned?	Reason	Person	Verified by (Optional)
Farmworker Main Number Telephony EP	FarmworkerMainMenu	9640	Abandoned	Left Menu	Adam	Meera
Farmworker Main Number Telephony EP	LanguageSelectionMenu	156	Abandoned	Left Menu	Adam	Meera
Farmworker Main Number Telephony EP	MigrantVoicemailTransfer	67	Not Abandoned	Voicemail Transfer	Adam	Meera
Farmworker Main Number Telephony EP	DisconnectContact1	44	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	ClosedMenu	29	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	ClinicVoicemailTransfer	7	Not Abandoned	Voicemail Transfer	Adam	Meera
Farmworker Main Number Telephony EP	StaffDirectorySpanishTransfer	5	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	StaffDirectoryEnglishTransfer	3	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	AddressFaxHoursMenu	1	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	No Activity	1	Excluded	Undetermined	Meera	
Legal Benefits Menu Telephony EP	ClosedQueueMenu	4674	Not Abandoned	Closed Queue / Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	ClinicVoicemailTransfer	818	Not Abandoned	Voicemail Transfer	Marcos	Adam
Legal Benefits Menu Telephony EP	LegalServerScreenPop	715	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	DisconnectContact1	334	Not Abandoned	Closed Queue / Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	BenefitsMenu	294	Abandoned	Left Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	VeteransBenefitsVoicemailTransfer	185	Not Abandoned	Voicemail Transfer	Marcos	Adam
Legal Benefits Menu Telephony EP	QueueMenu1	73	Abandoned	Left Queue	Marcos	Adam
Legal Benefits Menu Telephony EP	PlayMOH300s	41	Abandoned	Left Queue	Marcos	Adam
Legal Benefits Menu Telephony EP	ScreenPopProcessComplete	41	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	IntakePreQueueMessage1	17	Abandoned	Left Queue	Marcos	Adam
Legal Benefits Menu Telephony EP	DisconnectCallbackContact	8	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	CallbackRetry	3	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	LegalMenu2	3	Abandoned	Left Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	ClosedMenu	1	Not Abandoned	Closed Queue / Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	ConfirmCallbackNumber	1	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	PreQueueMessage2	1	Abandoned	Left Queue	Marcos	Adam
Legal Benefits Menu Telephony EP	StaffDirectoryEnglishTransfer	1	Not Abandoned	Closed Queue / Menu	Marcos	Adam
Legal Employment Menu Telephony EP	ClosedQueueMenu	3063	Not Abandoned	Closed Queue / Menu	Adam	Meera
Legal Employment Menu Telephony EP	ClinicVoicemailTransfer	536	Not Abandoned	Voicemail Transfer	Adam	Meera
Legal Employment Menu Telephony EP	LegalServerScreenPop	502	Not Abandoned	Callback (or Callback Attempt)	Adam	Meera
Legal Employment Menu Telephony EP	EmploymentMenu	182	Abandoned	Left Menu	Adam	Meera
Legal Employment Menu Telephony EP	DisconnectContact1	137	Not Abandoned	Closed Queue / Menu	Adam	Meera
Legal Employment Menu Telephony EP	QueueMenu1	33	Abandoned	Left Queue	Adam	Meera
Legal Employment Menu Telephony EP	ScreenPopProcessComplete	31	Not Abandoned	Callback (or Callback Attempt)	Adam	Meera

Common Reasons (to copy paste, for consistency)

Abandoned	Left Menu
Abandoned	Left Queue
Not Abandoned	Closed Queue / Menu
Not Abandoned	Callback (or Callback Attempt)
Not Abandoned	Voicemail Transfer
Not Abandoned	Call Ended
Not Abandoned	Not Served by Legal Aid
Not Abandoned	Transfer
Not Abandoned	Call Concluded
Not Abandoned	Received Info

Classifying Unclear Subgroups

Menu	Last Activity	Abandoned?	Reason	Person	Verified By
Legal Housing Menu Telpephony EP	StaffDirectoryEnglishTransfer	?	?	?	?



```
# 1. Find the Contact Session IDs that match the criteria
family_ids = df_final.loc[
    (df_final["Menu"] == "Legal Housing Menu Telephony EP") &
    (df_final["Last Activity Name"] == "StaffDirectoryEnglishTransfer"),
    "Contact Session ID"
].unique()

# 2. Pull ALL rows from df_main for these Contact Session IDs
test_output = df_main[df_main["Contact Session ID"].isin(family_ids)].copy()

test_output.head(200)
```



d44b132c-e870-4cd2-bc88-68ae441f6625	NaN	ClosedHoursHolidaysMenu	NaN	2024-05-02 16:34:17	NaN	NaN
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	ClosedMenu	2024-05-02 16:34:17	NaN	NaN
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	StaffDirectoryEnglishTransfer	2024-05-02 16:34:18	NaN	NaN
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	2024-05-02 16:34:18	NaN	NaN
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	2024-05-02 16:34:18	Staff Directory English Transfer	CBT Agent
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	2024-05-02 16:35:03	Staff Directory English Transfer	CBT Agent
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	2024-05-02 16:35:03	Staff Directory English Transfer	CBT Agent



Menu	Last Activity	Abandoned?	Reason	Person	Verified By
Legal Housing Menu Telpephony EP	StaffDirectoryEnglishTransfer	Not Abandoned	Call Concluded	Adam	Meera

Abandonment Results (Marcos)

Overall Call Abandonment Rate: 32.31%

Abandonment Rate by Menu, ordered by Abandonment Rate:

Menu Name	Total Calls	% of All Calls	Abandoned Calls	Abandonment Rate
Other Legal	12,204	5.33%	6,830	55.97%
Legal	27,715	12.11%	13,793	49.77%
HIV	1,603	0.70%	767	47.85%
Pre-Legal Seniors	12,939	5.65%	4,425	34.20%
Main	119,927	52.39%	40,657	33.90%
Housing	16,492	7.20%	3,837	23.27%
Immigration	3,516	1.54%	696	19.80%
Family	22,753	9.94%	2,242	9.85%
Employment	4,564	1.99%	289	6.33%
Benefits	7,210	3.15%	429	5.95%

Takeaways (Sana)

- **Abandonment is extremely high vs. norms**

Across the 228,923 calls that hit one of the menus, **73,965 were abandoned**, an overall abandonment rate of **32.3%**, which is **4–6x higher** than the typical call-center benchmark of **5–8%**.

- **Most abandoned calls are concentrated in a few menus**

Just **five menus** — **Main, Legal, Other Legal, Pre-Legal Seniors, and Housing** — **account for ~94% of all abandoned calls**. The Main menu alone contributes about **55%** of all abandonments (40,657 calls), followed by Legal (18.6%) and Other Legal (9.2%).

- **Several menus have abandonment rates that are alarmingly high**

Compared with the 5–8% benchmark, abandonment is particularly severe in **Other Legal (56%), Legal (50%), HIV (48%), Pre-Legal Seniors (34%), Main (34%), Housing (23%), and Immigration (20%)**.

- **“Low” abandonment menus are not automatically success stories**

Menus like **Benefits (6.0%), Employment (6.3%), and Family (9.9%)** are near or only slightly above industry norms, and may offer useful models for routing or scripting. But even in these menus, some callers still hit closed queues or hang up before service, so “low” abandonment does not necessarily mean the system is fully meeting demand.

Abandonment: Queues vs. Menus (Sana)

Menu Name	Total Abandoned Calls	Abandoned in Menu	Abandonment in Queue
Benefits	429	297	132
Employment	289	214	75
Family	2242	1874	368
HIV	767	767	0 (No queues)
Housing	3,837	3608	229
Immigration	696	641	55
Legal	13,793	13653	140
Main	40,657	40657	0 (No queues)
Other Legal	6,830	6830	0 (No queues)
Pre-Legal Seniors	4,425	2809	1616

Abandonment: Queues vs. Menus (Sana)

Looking closer at the distribution between abandonment within queues and menus, some menus had greater disparities than others.

Housing Menu: Of the 3,837 calls that abandoned in the housing menu, 94% ended within a submenu of the housing menu. The majority of abandonment was within the **Tenant Menu**, at 2,394 calls.

Legal Menu: Of the 13,793 calls abandoned in the legal menu, 99% ended within a submenu of the legal menu. The majority of the abandonment cases was in **Legal Menu 2** at 11,969 calls.

Family Menu: Of the 2,242 call abandoned in the family menu, 84% ended within a submenu of the family menu. The majority of the abandoned cases was in the **Divorce Or Parenting Menu** at 1,462 calls.

While higher menu-level abandonment is likely driven by the amount of navigation required within these menus, the specific submenus where callers most frequently drop off can point to particular wording or instructions that may be unclear. These patterns highlight where refining verbiage or simplifying the path could significantly improve caller clarity and reduce abandonment.

Hidden Barriers in “Low–Abandonment” Menus

Even menus with low abandonment rates still show **very high rates of closed queue/menu outcomes**, revealing that callers often cannot access live assistance even when they stay in the system.

Benefits Menu:

Out of 7,210 total calls, **5,010** ended in a closed queue/menu — a **69% closed rate**.

Employment Menu:

Out of 4,564 total calls, **3,201** were closed queue/menu — a **70% closed rate**.

Family Menu:

Out of 22,753 total calls, **14,344** were closed queue/menu — a **63% closed rate**.

Low abandonment does **not** mean the menu is performing well — it often masks the fact that callers are being routed into **closed or unavailable queues** rather than receiving assistance.

Recommendations (Marcos)

- **Set a concrete abandonment-reduction goal and focus on the highest-impact menus first**
As a first phase, aim to cut overall abandonment from **32% toward ~15%**, prioritizing **Main, Legal, Other Legal, Pre-Legal Seniors, and Housing**, since they drive almost all abandoned calls. Even modest improvements in these menus will materially improve access.
- **Simplify and speed up the Main menu experience**
Since over half of all abandoned calls die in the Main menu, review menu length, wording, and branching.
- **Address capacity and routing issues in Legal / Other Legal / Pre-Legal Seniors**
The near-50% abandonment in Legal and Other Legal suggests callers are either waiting too long, looping between menus, or hitting closed/overloaded queues. Consider increasing staffing or redistributing agents during peak hours and creating a dedicated, simpler path for seniors and high-priority legal issues.
- **Conduct targeted qualitative reviews for high-rate, lower-volume menus (HIV, Immigration)**
Because these menus have fewer calls but very high abandonment, a small number of design issues could be driving outsized drop-off. Sample call listening and journey mapping can identify whether callers are dropping due to confusing language, stigma-related concerns, or long silences before reaching a person.

Works Cited

Python Code (Adam) (to be added):

R Code (Meera):

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/Meera%20Trivedi/Total_Abandonment_4Dec_Meera.R

Spreadsheet:

<https://docs.google.com/spreadsheets/d/1z8cZ-0Hxflcaw1d8fS9jQddN1noD1cVIIUNTUEO93ul/edit?gid=0#gid=0>